Warden’s Key: Cryptography

Creator: Ammani Khan Points: –

Description

Competitors will decrypt a classical Hill cipher using a 2×2 matrix key. The challenge tests understanding of modular arithmetic, matrix operations, and the ability to reverse linear transformations in cryptographic systems.

Github Path: Cryptography/Wardens_Key

Prompt

In the depths of an ancient forest, you stumble upon the ruins of a forgotten watchtower. Vines snake through crumbled stone, and inside, you find a dusty journal half-buried beneath fallen leaves.

The journal belonged to a forest warden who once guarded these woods. On the page, a scrawled note with a key to pass through:

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

Encrypted Message: OSWPLPLDUY

This mathematical lock is used by wardens to protect secret passages. Decrypt the message to discover the word that unlocks the passage.

Format: flame{plaintext}

Hints

1. The Hill cipher works on pairs of letters. Split the ciphertext into 2-letter blocks.
2. To decrypt, you need the inverse of the key matrix modulo 26.
3. Remember: det(K) = ad - bc. Find the modular multiplicative inverse of the determinant mod 26.

Solution

1. Split ciphertext into pairs (OS, WP, LP, LD, UY)
2. Calculate determinant det(K) = (3 × 5) - (3 × 2) = 9
   Find modular inverse of 9 mod 26. 9 × 3 = 27 ≡ 1 (mod 26)
   So $9^{-1} = 3$
3. Calculate adjugate matrix
   adj(K) = [5, -3; -2, 3]
4. Calculate inverse matrix
   $K^{-1} = 3 \times [5, -3; -2, 3] \bmod 26$
   $K^{-1} = [15, -9; -6, 9] \bmod 26$
   $K^{-1} = [15, 17; 20, 9]$
5. Decrypt each pair
   OS = (14, 18) → WA = (22, 0)
   WP = (22, 15) → ND = (13, 3)
   LP = (11, 15) → ER = (4, 17)

```
      LD = (11, 3) → IN = (8, 13)
      UY = (20, 24) → GS = (6, 18)
    6.  Final answer: WANDERINGS
```

Flag

```
flame{wanderings}
```